# Customer Shopping Behavior Analysis

**Created by Anjul Sharma**

---

# 1. Project Overview

This project analyzes customer shopping behavior using transactional data from **3,900 purchases** across different product categories. The objective of this analysis is to understand spending patterns, customer segments, product preferences, and subscription behavior.

The project uses **Python for data cleaning**, **SQL for business analysis**, and **Power BI for dashboard visualization** to generate meaningful business insights.

---

# 2. Dataset Summary

The dataset contains transactional data related to customer purchases.

**Dataset Details**

- Rows: 3,900
- Columns: 18

**Key Features**

Customer Demographics

- Age
- Gender
- Location
- Subscription Status

Purchase Details

- Item Purchased
- Category
- Purchase Amount
- Season
- Size
- Color

Shopping Behavior

- Discount Applied

- Previous Purchases
- Frequency of Purchases
- Review Rating
- Shipping Type

**Missing Data**

The *Review Rating* column contained missing values which were handled using **category-wise median imputation**.

---

# 3. Exploratory Data Analysis using Python

Data cleaning and preparation were performed using Python and the Pandas library.

## Data Loading

The dataset was loaded using Pandas.

```
import pandas as pd
df = pd.read_csv("customer_shopping_behavior.csv")
```

---

## Initial Data Exploration

The structure and statistical summary of the dataset were examined.

```
df.info()
df.describe()
df.isnull().sum()
```

---

## Missing Value Handling

Missing values in the **Review Rating** column were filled using the median rating of each product category.

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x:
x.fillna(x.median()))
```

---

## Column Standardization

Column names were standardized using snake_case formatting.

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ','_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

---

### Feature Engineering

Two new features were created.

#### Age Group Segmentation

Customers were divided into age groups.

```
labels = ['Young Adult','Adult','Middle-aged','Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)
```

---

#### Purchase Frequency in Days

Purchase frequency values were converted into numeric days.

```
frequemcy_mapping = {
'Fortnightly':14,
'Weekly':7,
'Monthly':30,
'Quarterly':90,
'Bi-Weekly':14,
'Annually':365,
'Every 3 Months':90
}

df['purchse_frequency_days'] = df['frequency_of_purchases'].map(frequemcy_mapping)
```

---

### Data Consistency Check

The column `promo_code_used` was removed because it provided redundant information.

```
df = df.drop('promo_code_used',axis=1)
```

---

### Database Integration

The cleaned dataset was loaded into a MySQL database for SQL analysis.

```
from sqlalchemy import create_engine

engine = create_engine("mysql+pymysql://root:anjul@localhost:3306/mydatabase")
df.to_sql("mytable", engine, if_exists="replace", index=False)
```

---

# 4. Data Analysis using SQL (Business Transactions)

SQL queries were used to answer important business questions.

# 1. Revenue by Gender

This analysis compares the total revenue generated by male and female customers.

Insight:
Male customers generated significantly higher revenue compared to female customers.

```
-- Q1. What is the total revenue generated by male vs. female customers?

select gender ,sum(purchase_amount) as revenue
from mytable
group by gender;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| gender | revenue |
|---|---|
| Male | 157890 |
| Female | 75191 |

# 2. High-Spending Discount Users

Customers who used discounts but still spent above the average purchase amount were identified.

Insight:
Some high-value customers still use discounts, showing that promotional strategies attract both low and high spenders.

```sql

-- Q2. Which customers used a discount but still spent more than the average purchase amount?

select customer_id,purchase_amount
from mytable
where discount_applied = 'Yes' and purchase_amount >= (select avg(purchase_amount) from mytable);

```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| customer_id | purchase_amount |
|---|---|
| 2 | 64 |
| 3 | 73 |
| 4 | 90 |
| 7 | 85 |
| 9 | 97 |
| 12 | 68 |
| 13 | 72 |
| 16 | 81 |
| 20 | 90 |

---

# 3. Subscribers vs Non-Subscribers

This analysis compares the spending behavior of subscribed and non-subscribed customers.

Insight:
Average spending between subscribers and non-subscribers is similar, but non-subscribers contribute more overall revenue due to a larger customer base.

```sql

-- Q5. Do subscribed customers spend more? Compare average spend and total revenue between subscribers and non-subscribers.
SELECT
    subscription_status,
    AVG(purchase_amount) AS avg_spend,
    SUM(purchase_amount) AS total_revenue
FROM mytable
GROUP BY subscription_status;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| subscription_status | avg_spend | total_revenue |
|---|---|---|
| Yes | 59.4919 | 62645 |
| No | 59.8651 | 170436 |

---

# 4. Customer Segmentation

Customers were segmented based on the number of previous purchases.

Segments:

- New Customers
- Returning Customers
- Loyal Customers

Insight:
Loyal customers represent the largest high-value segment and contribute significantly to revenue.

```sql
-- Q7. Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.
WITH customer_filtered AS (
    SELECT
        customer_id,
        previous_purchases,
        purchase_amount,
        CASE
            WHEN previous_purchases = 1 THEN 'New'
            WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| customer_segment | number_of_customers | avg_spend |
|---|---|---|
| Returning | 143 | 79.83 |
| Loyal | 681 | 79.84 |
| New | 15 | 76.93 |

---

# 5. Revenue by Age Group

This analysis identifies which age group contributes the most revenue.

Insight:
Young Adult customers generated the highest revenue compared to other age groups.

```sql
-- Q10. What is the revenue contribution of each age group?
SELECT
    age_group,
    SUM(purchase_amount) AS total_revenue
FROM mytable
GROUP BY age_group
ORDER BY total_revenue DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

| age_group | total_revenue |
|---|---|
| Young Adult | 62143 |
| Middle-aged | 59197 |
| Adult | 55978 |
| Senior | 55763 |

# 5. Dashboard in Power BI

An interactive Power BI dashboard was created to visualize key insights.

The dashboard includes:

- Total Customers KPI
- Average Purchase Amount
- Average Review Rating
- Revenue by Category
- Revenue by Age Group
- Subscription Distribution

The dashboard allows users to filter and analyze customer behavior dynamically.

# 6. Business Recommendations

Based on the analysis, the following recommendations are suggested:

- Promote subscription benefits to increase subscriber base
- Implement loyalty programs for repeat customers
- Optimize discount strategies to improve profit margins
- Focus marketing campaigns on high revenue age groups
- Promote high-performing product categories